

ORF309/MAT380

Random Walks III: Biased Random Walks

(pp. 127-133)

Ioannis Akrotirianakis

Agenda for today

- **What happens when life is not fair (i.e., when there is bias)**

- $f(i) = P\{T = \infty | S_0 = i\} = ?$

- $r(i) = P\{S_T = b | S_0 = i\} = ?$

- $\mu(i) = E[T | S_0 = i] = ?$

- Gambler's ruin ?

Recap on the definitions of RWs

- **Definition:** A **random walk** $\{S_n\}_{n \geq 0}$ is a random process whose k -th element is given by $S_k = S_0 + X_1 + X_2 + \dots + X_k$, where $X_1, X_2, \dots$ are **i.i.d.** random variables **also independent** of the initial state S_0 .
- **Definition:** A random walk $\{S_n\}_{n \geq 0}$ is called **simple** if S_0 is integer and the steps X_k can only take ± 1 values.
- **Definition:** A random walk $\{S_n\}_{n \geq 0}$ is called **symmetric** if it is **simple** and

$$P\{X_k = 1\} = P\{X_k = -1\} = 1/2$$

- **Expectations:** $E[X_k] = 1 \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0$, $E[S_n | S_0 = i] = i + \sum_{k=1}^n E[X_k] = i \rightarrow$ **Fair game**

Biased Random Walks

Biased RWs

- In **Biased RWs** the steps $X_1, X_2, X_3, \dots$ are **still i.i.d. and independent of S_0** **but**

$$P\{X_k = 1\} = p \text{ and } P\{X_k = -1\} = 1 - p = q \text{ with } p + q = 1$$

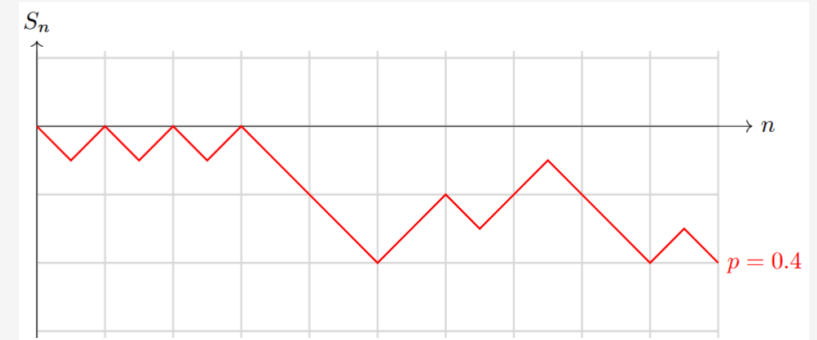
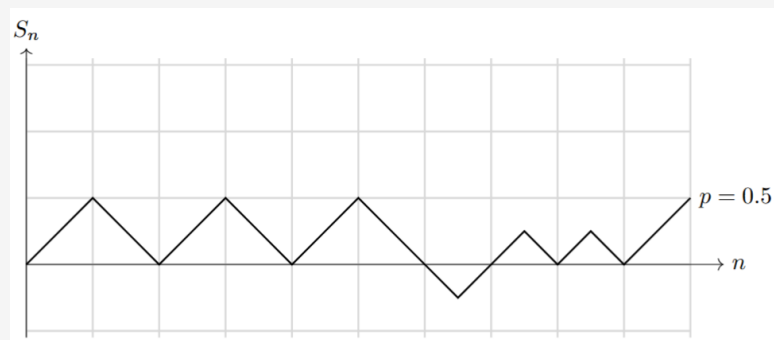
- The **Expectations** then become:

$$E[X_k] = \boxed{??}$$

and

$$E[S_n | S_0 = i] =$$

$\boxed{??}$



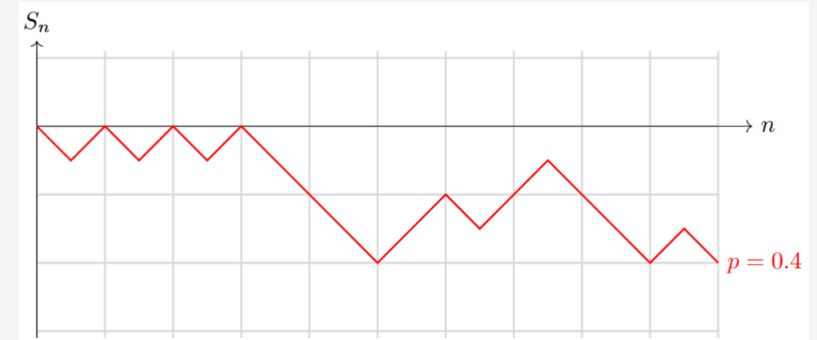
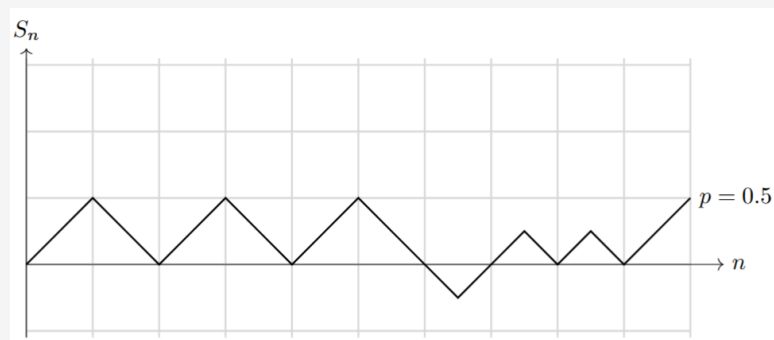
Biased RWs

- In **Biased RWs** the steps $X_1, X_2, X_3, \dots$ are **still i.i.d. and independent of S_0** **but**

$$P\{X_k = 1\} = p \text{ and } P\{X_k = -1\} = 1 - p = q \text{ with } p + q = 1$$

- The **Expectations** then become:

$$E[X_k] = p - q \text{ and } E[S_n | S_0 = i] = i + \sum_{k=1}^n E[X_k] = i + n(p - q) \begin{matrix} > i \\ < i \end{matrix}$$



-
- **Hitting time** is defined exactly the same way

$$T = \min\{n : S_n = a \text{ or } S_n = b\}$$

- We **still** have to answer the **same 3 questions**
 - **Question 1:** will the game end (hit the barrier), or go on for ever?
 - **Question 2:** if it ends, what is the probability you will reach your goal?
 - **Question 3:** on average, how long will it take for the game to end?

Q1: Will we ever hit the boundary?

- We define the function:

$$f(i) = P\{T = \infty | S_0 = i\}$$

- Working **similarly** as in the symmetric RW and using First Step Analysis case we get

$$f(i) = pf(i+1) + qf(i-1), \quad a < i < b$$

$$p + q = 1$$

$$f(a) = 0, \quad f(b) = 0$$

- If $p = q = 1/2$ then we easily can show: $f(i) = P\{T = \infty | S_0 = i\} = 0, \quad \forall a \leq i \leq b$
- But for the **biased case** we have to be more careful...

-
- We will use the fact: $p + q = 1$

$$f(i) = pf(i+1) + qf(i-1) \Leftrightarrow$$

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$$\Leftrightarrow p(f(i+1) - f(i)) = q(f(i) - f(i-1)) \Leftrightarrow f(i+1) - f(i) = \frac{q}{p} (f(i) - f(i-1))$$

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- We know: $f(a) = 0$; and we define: $f(a+1) - f(a) = u$; and then compute $(i-1)$ equations:

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- We know: $f(a) = 0$; and we define: $f(a+1) - f(a) = u$; and then compute $(i-1)$ equations:

$$i \leftarrow a + 1 \Rightarrow f(a+2) - f(a+1) = \frac{q}{p} (f(a+1) - f(a)) = \left(\frac{q}{p}\right)^1 u$$

$$i \leftarrow a + 2 \Rightarrow f(a+3) - f(a+2) = \frac{q}{p} (f(a+2) - f(a+1)) = \left(\frac{q}{p}\right)^2 u$$

...

...

$$i \leftarrow a + i - 1 \Rightarrow f(a+i) - f(a+i-1) = \frac{q}{p} (f(a+i-1) - f(a+i-2)) = \left(\frac{q}{p}\right)^{i-1} u$$

-
- Adding up the previous $i + 1$ equations (**including:** $f(a) = 0$ **and** $f(a + 1) - f(a) = \mathbf{u}$), yields:

$$f(a + i) = \left(1 + \frac{q}{p} + \left(\frac{q}{p} \right)^2 + \cdots + \left(\frac{q}{p} \right)^{i-1} \right) \mathbf{u} =$$

-
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$$f(a + i) = \overbrace{\left(1 + \frac{q}{p} + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{i-1} \right)}^{\text{Geometric sequence}} u = \frac{\left(\frac{q}{p}\right)^i - 1}{\frac{q}{p} - 1} u \quad u \text{ is still unknown}$$

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- But we **know** that $f(b) = 0$, which can help us compute u

$$f(b) = 0 \Leftrightarrow f(a + b - a) = 0 \Leftrightarrow \frac{\left(\frac{q}{p}\right)^{b-a} - 1}{\left(\frac{q}{p} - 1\right)} u = 0 \Leftrightarrow \mathbf{u = 0}$$

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Hence: $f(a + i) = 0$, $\forall i$, and equivalently: $f(i) = \mathbf{P}\{T = \infty | S_0 = i\} = \mathbf{0}$, $a \leq i \leq b$

Q2: Which boundary will we hit?

- We define the probability of hitting b before hitting a :

$$r(i) = P\{S_T = b | S_0 = i\}$$

- Using First Step Analysis (exactly as in **Q1**) we obtain the system:

$$r(i) = pr(i+1) + qr(i-1), \quad a < i < b$$

$$r(a) = 0, \quad r(b) = 1$$

- We will work similarly as for $f(i)$

-
- We can use the fact: $p + q = 1$ and show:

$$r(i + 1) - r(i) = \frac{q}{p} (r(i) - r(i - 1))$$

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- We **know**: $r(a) = 0$; and we **define**: $r(a + 1) - r(a) = u$; and then **compute** the $(i - 1)$ equations:

$$i \leftarrow a + 1 \Rightarrow r(a + 2) - r(a + 1) = \frac{q}{p} (r(a + 1) - r(a)) = \left(\frac{q}{p}\right)^1 u$$

$$i \leftarrow a + 2 \Rightarrow r(a + 3) - r(a + 2) = \frac{q}{p} (r(a + 2) - r(a + 1)) = \dots = \left(\frac{q}{p}\right)^2 u$$

... ..

$$i \leftarrow a + i - 1 \Rightarrow r(a + i) - r(a + i - 1) = \dots = \left(\frac{q}{p}\right)^{i-1} u$$

-
- Adding up the previous **$i + 1$** equations (**including** $r(a) = 0$ **and** $r(a + 1) - r(a) = u$) yields:

$$r(a + i) = \left(1 + \frac{q}{p} + \left(\frac{q}{p} \right)^2 + \cdots + \left(\frac{q}{p} \right)^{i-1} \right) u = \frac{\left(\frac{q}{p} \right)^i - 1}{\left(\frac{q}{p} - 1 \right)} u$$

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- But now we have $r(b) = 1$

$$r(b) = 1 \Leftrightarrow r(a + b - a) = 1 \Leftrightarrow \frac{\left(\frac{q}{p} \right)^{b-a} - 1}{\left(\frac{q}{p} - 1 \right)} u = 1 \Leftrightarrow$$

-
- Adding up the previous $i + 1$ equations (**including** $r(a) = 0$ **and** $r(a + 1) - r(a) = u$) yields:

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- But now we have $r(b) = 1$

$$r(b) = 1 \Leftrightarrow r(a + b - a) = 1 \Leftrightarrow \frac{\left(\frac{q}{p}\right)^{b-a} - 1}{\left(\frac{q}{p} - 1\right)} u = 1 \Leftrightarrow u = \frac{\left(\frac{q}{p} - 1\right)}{\left(\frac{q}{p}\right)^{b-a} - 1}$$

Hence: $r(a + i) = \frac{\left(\frac{q}{p}\right)^i - 1}{\left(\frac{q}{p}\right)^{b-a} - 1}, \quad \forall i,$

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- But now we have $r(b) = 1$

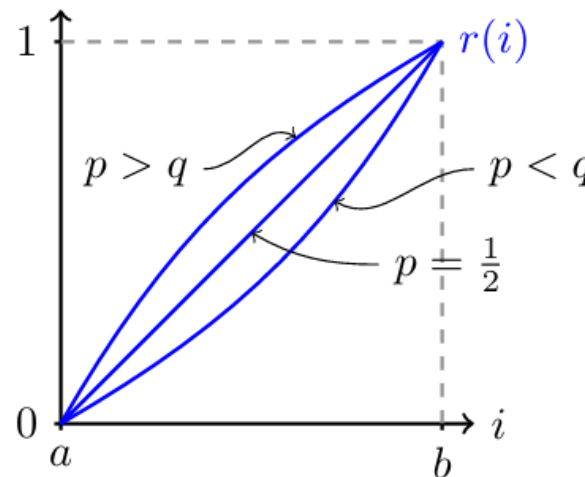
$$r(b) = 1 \Leftrightarrow r(a + b - a) = 1 \Leftrightarrow \frac{\left(\frac{q}{p}\right)^{b-a} - 1}{\left(\frac{q}{p} - 1\right)} u = 1 \Leftrightarrow u = \frac{\left(\frac{q}{p} - 1\right)}{\left(\frac{q}{p}\right)^{b-a} - 1}$$

Hence: $r(a + i) = \frac{\left(\frac{q}{p}\right)^i - 1}{\left(\frac{q}{p}\right)^{b-a} - 1}, \forall i,$ and equivalently: $r(i) = P\{S_T = b | S_0 = i\} = \frac{\left(\frac{q}{p}\right)^{i-a} - 1}{\left(\frac{q}{p}\right)^{b-a} - 1}, a \leq i \leq b$

- **Remark:** The previous formula $r(i)$ for biased RWs **cannot be reduced** to the corresponding formula for **symmetric RWs**, where $p = q = 1/2$. This is because if we substitute $p = q = 1/2$ we get $0/0$ which is not defined! But then we could use **ℓ' Hopital's formula** with the **limit $q \rightarrow 1/2$** , and the biased formula for $r(i)$ will converge to the one for the symmetric RWs.

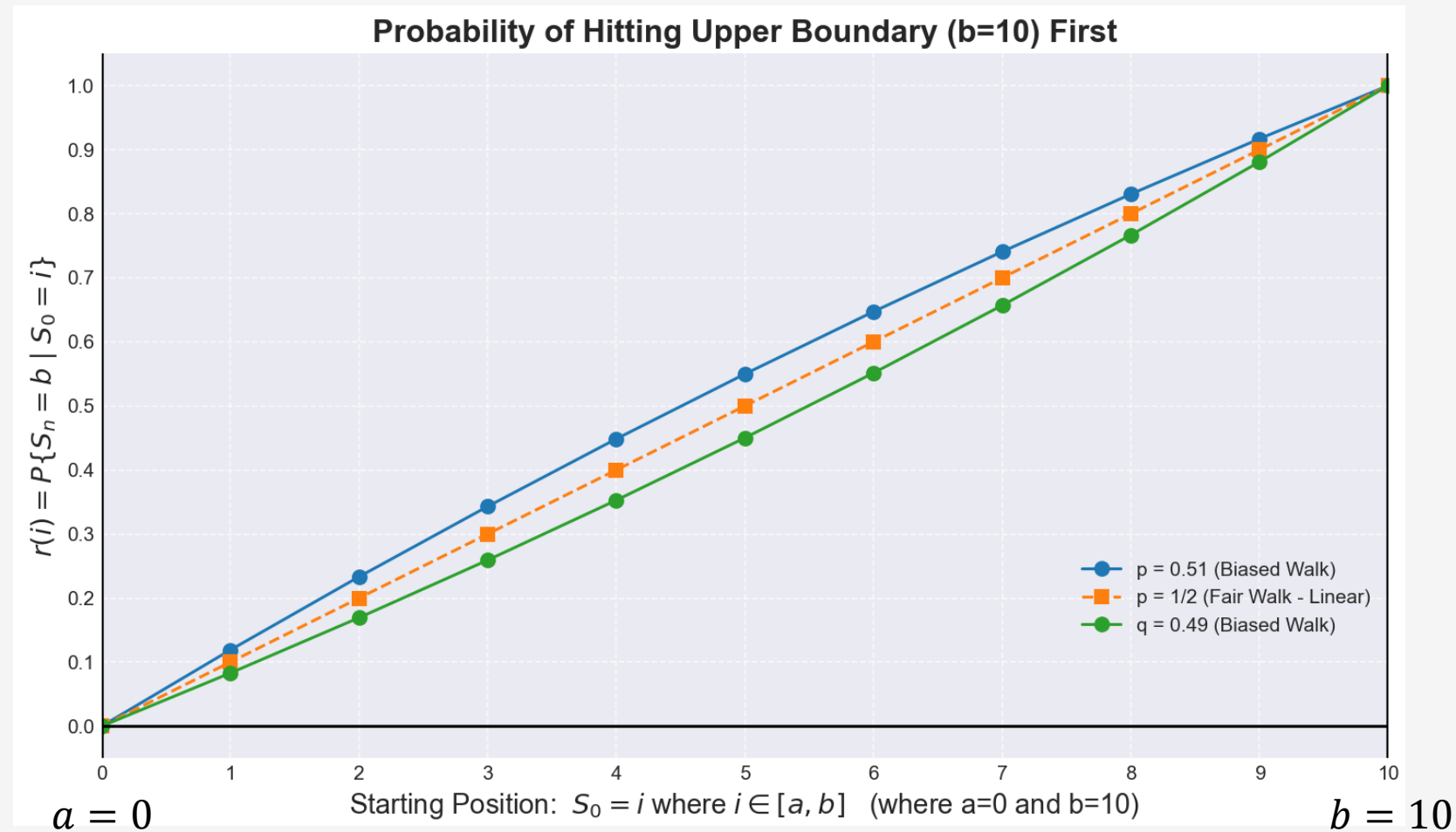
- **Remark:** the plot of biased $r(i)$ is:

If $p > q$:
You hit the " b " boundary **faster** than hitting the " a " boundary



If $p < q$:
You hit the " b " boundary **slower** than hitting the " a " boundary

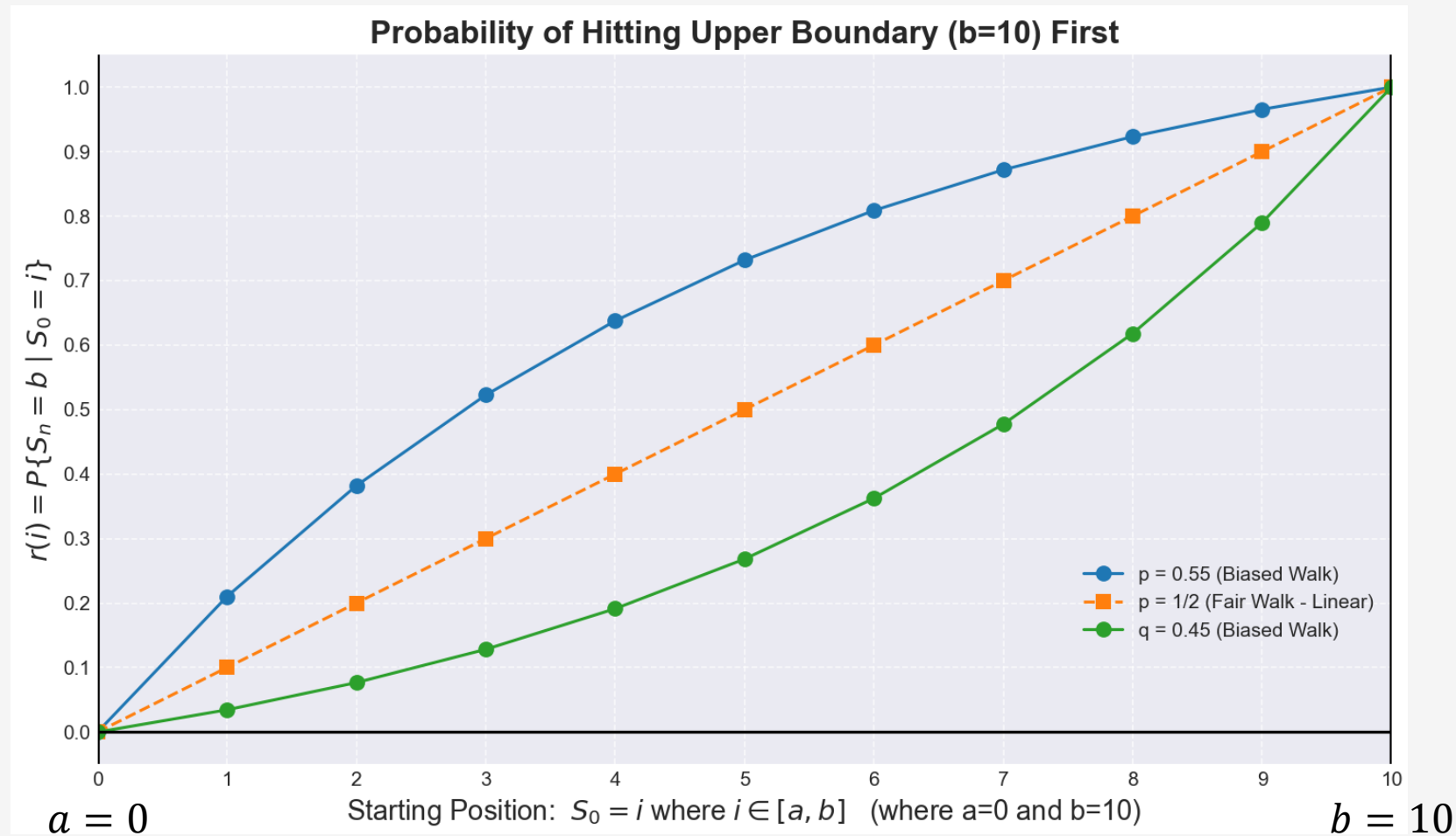
Illustrative simulations



The **blue** player has advantage over the **green** player

$r(i) =$	$r(i) =$
[−0.0,	[0.0,
0.1189,	0.083,
0.2332,	0.1693,
0.343,	0.2592,
0.4485,	0.3528,
0.5498,	0.4502,
0.6472,	0.5515,
0.7408,	0.657,
0.8307,	0.7668,
0.917,	0.8811,
1.0]	1.0]

Illustrative simulations

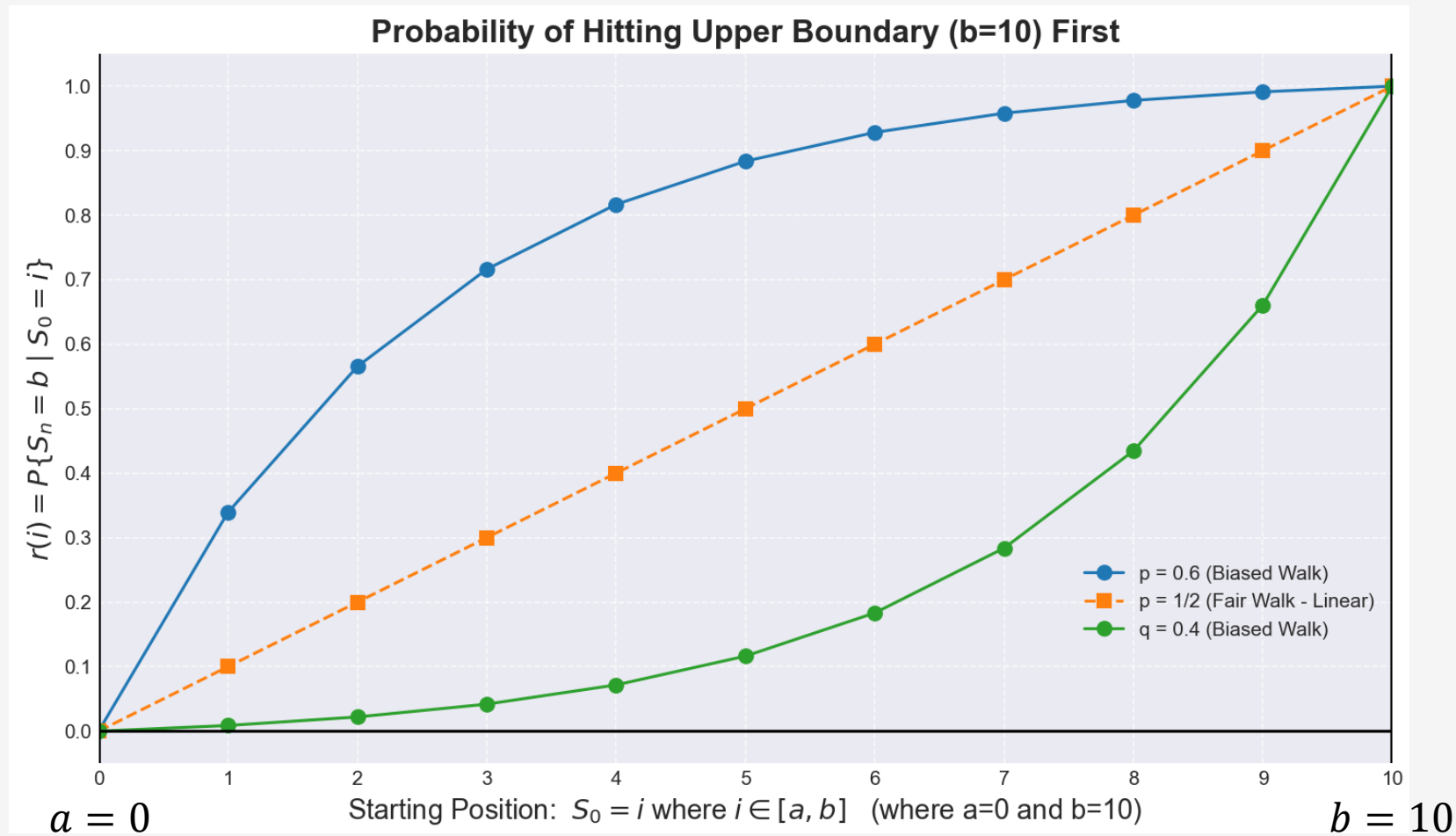


The **blue** player has advantage over the **green** player

$r(i) =$
 [0.0,
 0.2101,
 0.3819,
 0.5225,
 0.6376,
 0.7317,
 0.8087,
 0.8717,
 0.9233,
 0.9655,
 1.0]

$r(i) =$
 [0.0,
 0.0345,
 0.0767,
 0.1283,
 0.1913,
 0.2683,
 0.3624,
 0.4775,
 0.6181,
 0.7899,
 1.0]

Illustrative simulations



The **blue** player has advantage over the **green** player

$r(i) =$	$r(i) =$
[−0.0,	[0.0,
0.3392,	0.0088,
0.5654,	0.0221,
0.7161,	0.0419,
0.8166,	0.0717,
0.8836,	0.1164,
0.9283,	0.1834,
0.9581,	0.2839,
0.9779,	0.4346,
0.9912,	0.6608,
1.0]	1.0]

Q3: How long will it take to hit the boundary

- **Question:** on average, how long will it take for the game to end?
- To answer the question, we will compute the **Expected Hitting Time**, given that we start at position i :

$$T = \min\{n : S_n = a \text{ or } S_n = b\}$$

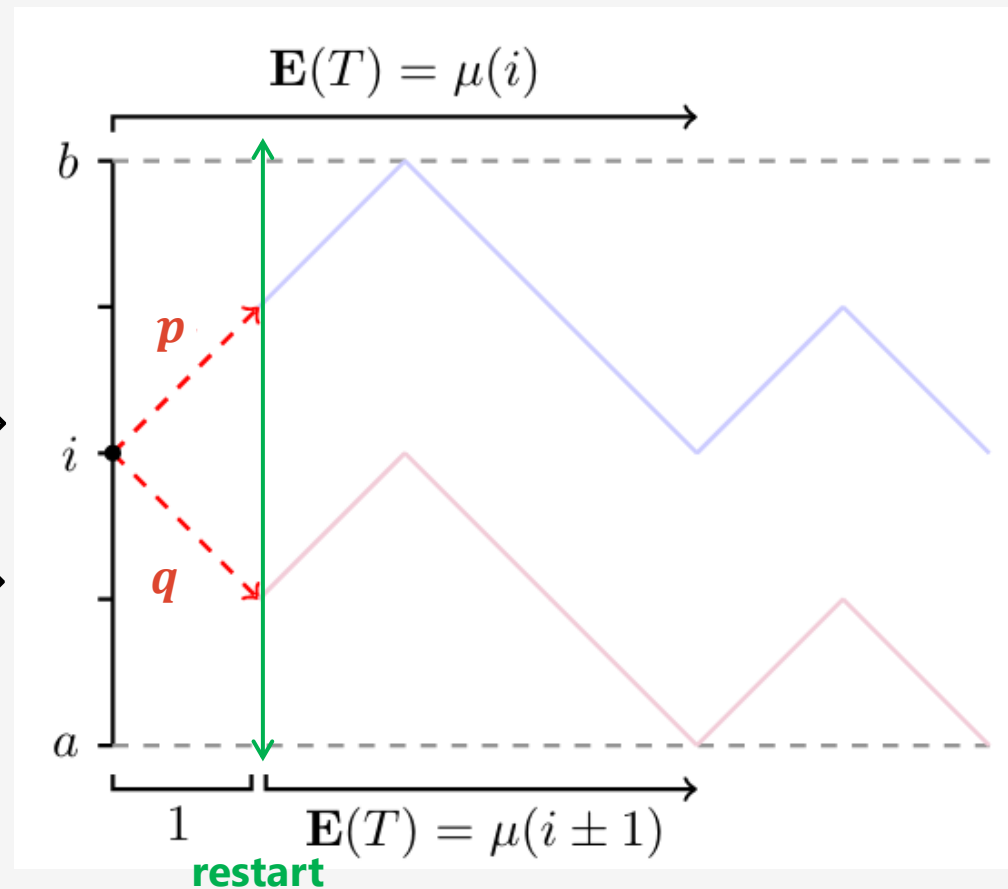
$$\mu(i) = E[T|S_0 = i], \quad a \leq i \leq b$$

- We will also use First Step Analysis to answer this question
- **If $i = a$ or $i = b$** then we are already on the **boundary**, so $T = 0$ and then:

$$\mu(A) = E[T|S_0 = i] = 0 \quad \text{and} \quad \mu(B) = E[T|S_0 = i] = 0$$

- If $a < i < b$ by First Step Analysis, we have:

$$\begin{aligned}
 \mu(i) &= E[T|S_0 = i] = \\
 &= (1 + E[T|S_0 = i + 1])P\{S_1 = i + 1|S_0 = i\} + \\
 &\quad \text{Going to } \mathbf{1}, \text{ and } \text{restarting} \\
 &+ (1 + E[T|S_0 = i - 1])P\{S_1 = i - 1|S_0 = i\} \Rightarrow \\
 &\Rightarrow \mu(i) = (1 + \mu(i + 1))\mathbf{p} + (1 + \mu(i - 1))\mathbf{q} \Rightarrow \\
 &\Rightarrow \mu(i) = 1 + \mathbf{p}\mu(i + 1) + \mathbf{q}\mu(i - 1)
 \end{aligned}$$



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- Hence, we have the system of linear equations:

$$\begin{cases} \mu(a) = 0, & \mu(b) = 0 \\ \mu(i) = 1 + p\mu(i+1) + q\mu(i-1), & a < i < b \end{cases} \quad \begin{matrix} (1) \\ (2) \end{matrix}$$

- We will use a more sophisticated trick...** We define:

$$\hat{\mu}(i) = \mu(i) + \frac{i-a}{p-q} \Rightarrow \mu(i) = \hat{\mu}(i) - \frac{i-a}{p-q} \quad (3)$$

- Substituting (2) into (3) yields:

$$\hat{\mu}(i) = 1 + p\mu(i+1) + q\mu(i-1) + \frac{i-a}{p-q} \quad (4)$$

-
- Evaluating $\mu(i)$ at $i + 1$ and $i - 1$ using (3) :

$$\mu(i + 1) = \hat{\mu}(i + 1) - \frac{i + 1 - a}{p - q}$$

$$\mu(i - 1) = \hat{\mu}(i - 1) - \frac{i - 1 - a}{p - q}$$

- ...and substituting into (4) yields:

$$\hat{\mu}(i) = 1 + p \left(\hat{\mu}(i + 1) - \frac{i + 1 - a}{p - q} \right) + q \left(\hat{\mu}(i - 1) - \frac{i - 1 - a}{p - q} \right) + \frac{i - a}{p - q} \Rightarrow \dots$$

$$\dots \Rightarrow \hat{\mu}(i) = p\hat{\mu}(i + 1) + q\hat{\mu}(i - 1), \quad a < i < b$$

-
- Let's see what happens on the boundary points a and b (use (3))

- $\hat{\mu}(a) = \mu(a) + \frac{a-a}{p-q} = 0 - 0 = 0$ and $\hat{\mu}(b) = \mu(b) + \frac{b-a}{p-q} = 0 + \frac{b-a}{p-q} = \frac{b-a}{p-q}$

- Hence, we have computed the **equivalent** system of equations:

$$\hat{\mu}(a) = 0, \quad \hat{\mu}(b) = \frac{b-a}{p-q}$$

$$\hat{\mu}(i) = p\hat{\mu}(i+1) + q\hat{\mu}(i-1), \quad a < i < b$$

This system of equations is the **same** as those for $r(i) = P\{S_T = b | S_0 = i\}$ (see slide 25). Hence:

$$\hat{\mu}(i) = \frac{(q/p)^{i-a} - 1}{q/p - 1} u, \quad (4) \quad (\text{where } u = \hat{\mu}(a+1) - \hat{\mu}(a))$$

$$\begin{aligned}\hat{\mu}(b) &= \frac{(q/p)^{b-a} - 1}{q/p - 1} u \Rightarrow \\ \frac{b-a}{p-q} &= \frac{(q/p)^{b-a} - 1}{q/p - 1} u \Rightarrow \\ u &= \frac{b-a}{p-q} \frac{q/p - 1}{(q/p)^{b-a} - 1}\end{aligned}$$

- To find the **unknown** u we use the **boundary** condition $\hat{\mu}(b)$ and substitute int (4):

$$\hat{\mu}(i) = \frac{(q/p)^{i-a} - 1}{(q/p)^{b-a} - 1} \cdot \frac{b-a}{p-q}, \quad a \leq i \leq b$$

Therefore, we have shown:

(see (3) in slide 32)

$$E[T|S_0 = i] = \mu(i) = \hat{\mu}(i) - \frac{i-a}{p-q} \Rightarrow \mu(i) = \frac{(q/p)^{i-a} - 1}{(q/p)^{b-a} - 1} \cdot \frac{b-a}{p-q} - \frac{i-a}{p-q}, \quad a \leq i \leq b$$

$$\begin{aligned}\hat{\mu}(b) &= \frac{(q/p)^{b-a} - 1}{q/p - 1} u \Rightarrow \\ \frac{b-a}{p-q} &= \frac{(q/p)^{b-a} - 1}{q/p - 1} u \Rightarrow \\ u &= \frac{b-a}{p-q} \frac{q/p - 1}{(q/p)^{b-a} - 1}\end{aligned}$$

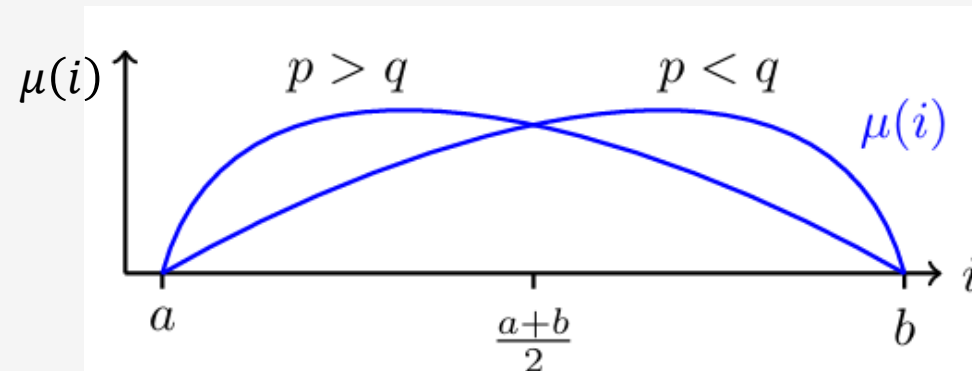
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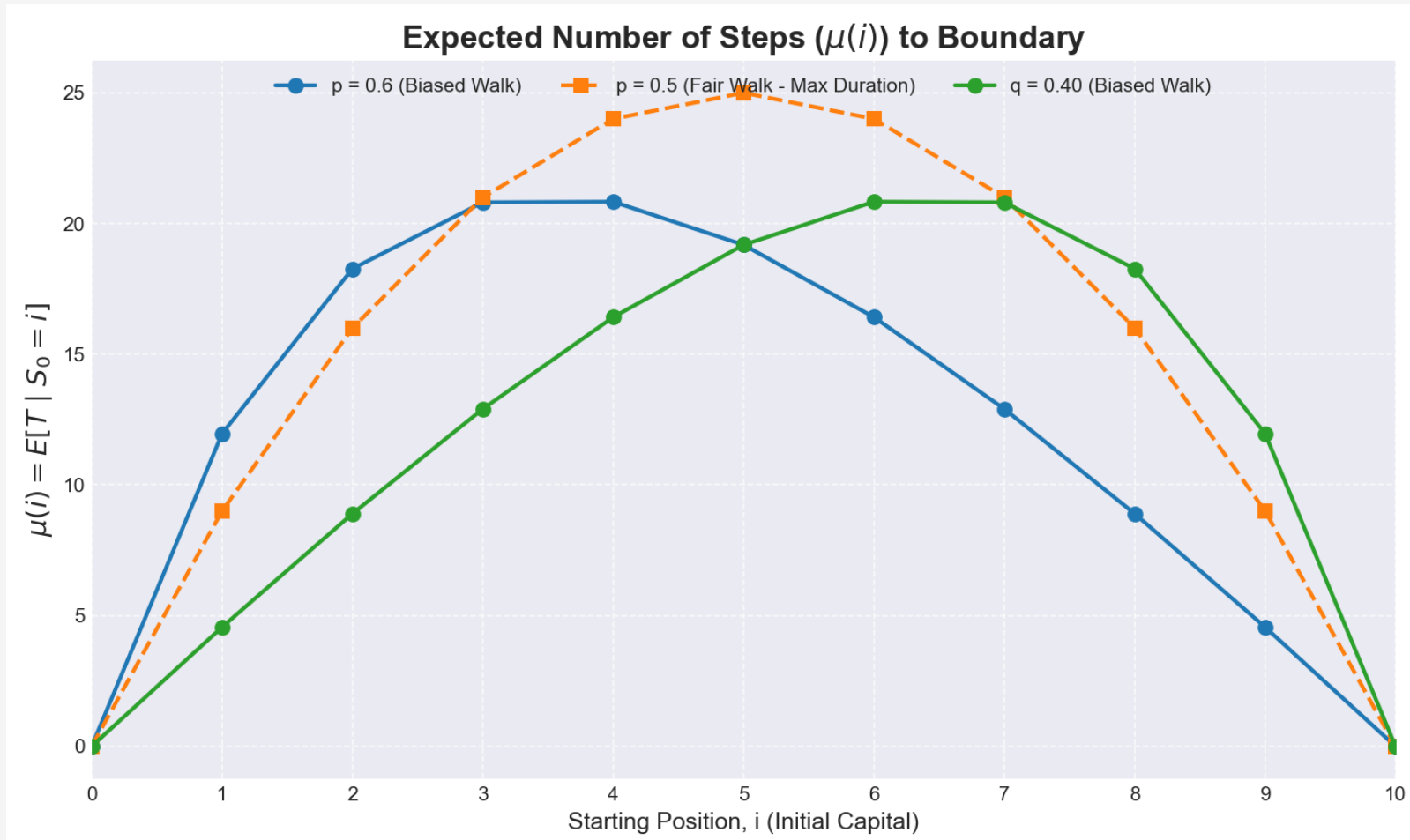
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(see (3) in slide 32)

$$E[T|S_0 = i] = \mu(i) = \hat{\mu}(i) - \frac{i-a}{p-q} \Rightarrow \mu(i) = \frac{(q/p)^{i-a} - 1}{(q/p)^{b-a} - 1} \cdot \frac{b-a}{p-q} - \frac{i-a}{p-q}, \quad a \leq i \leq b$$



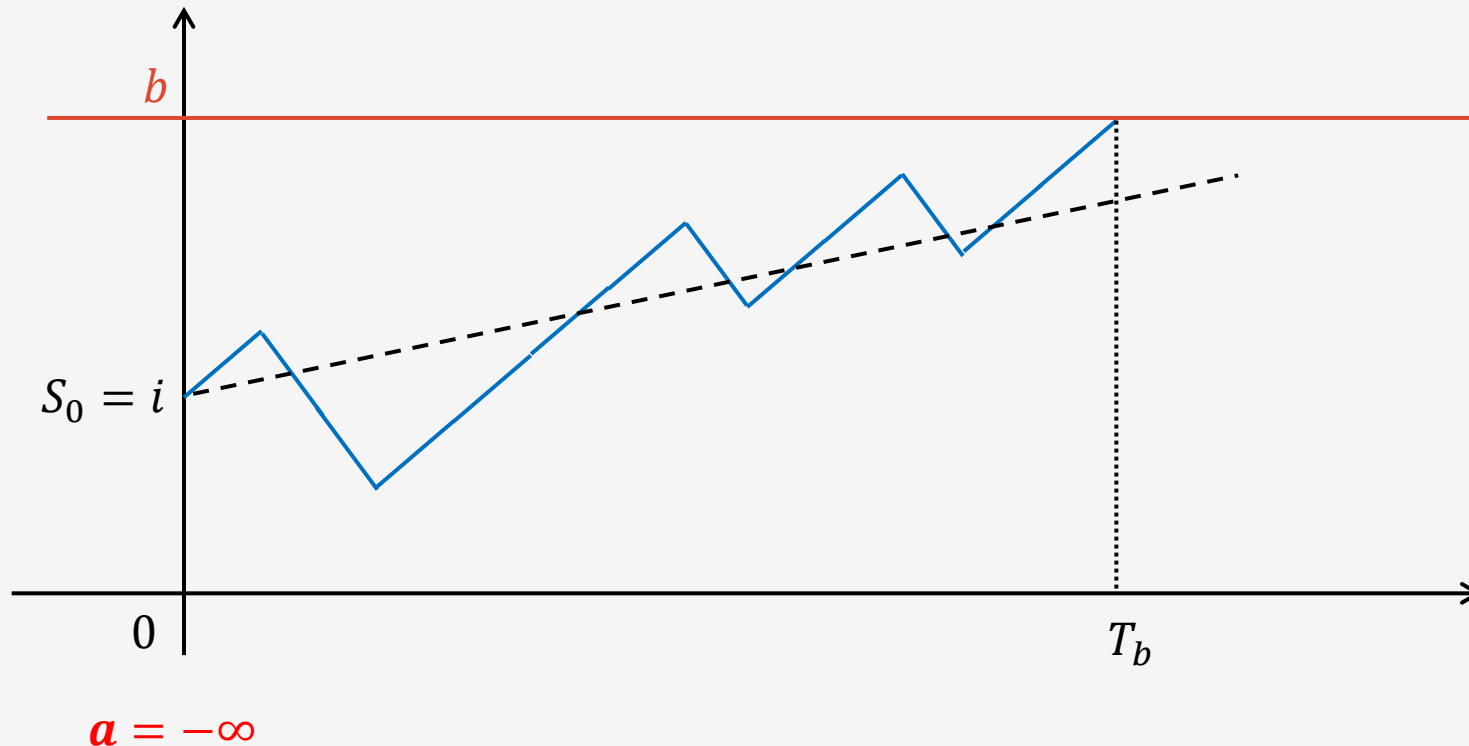
Illustrative simulations



Start	p=0.40	p=0.50	p=0.60
$S_0 = i$	$\mu(i)$	$\mu(i)$	$\mu(i)$
0	0.00	0.00	0.00
1	4.56	9.00	11.96
2	8.90	16.00	18.27
3	12.90	21.00	20.81
4	16.42	24.00	20.83
5	19.18	25.00	19.18
6	20.83	24.00	16.42
7	20.81	21.00	12.90
8	18.27	16.00	8.90
9	11.96	9.00	4.56
10	0.00	0.00	0.00

Gambler's ruin OR will we reach b ?

- Assume we have only an **upper barrier, b** (and **no lower barrier**, i.e., $a = -\infty$)
- What is the probability we hit b ? $\rightarrow P\{T_b < \infty | S_0 = i\} = ?$



- We can let $a \rightarrow \infty$ and apply the **two-boundary formula**:

$$P\{T_b < \infty | S_0 = i\} = \lim_{a \rightarrow -\infty} P\{S_T = b | S_0 = i\} = \lim_{a \rightarrow -\infty} \frac{(q/p)^{i-a} - 1}{(q/p)^{b-a} - 1}$$

$$T = \min \{n : S_n = a \text{ or } S_n = b\}$$

- We can distinguish **3 cases**:
- **Case #1**: $p = q = 1/2$: we have already proven that $P\{T_b < \infty | S_0 = i\} = 1$
- **Case #2**: $p > q \Rightarrow q/p < 1$ (on average we move more towards b (biased towards b))

$$\lim_{a \rightarrow -\infty} (q/p)^{i-a} = \lim_{a \rightarrow -\infty} (q/p)^{b-a} = 0$$

- Hence: $P\{T_b < \infty | S_0 = i\} = 1$

-
- **Case #3:** $p < q \Rightarrow q/p > 1$ (on average we move more towards a (biased towards a))

$$\lim_{a \rightarrow -\infty} (q/p)^{i-a} = \lim_{a \rightarrow -\infty} (q/p)^{b-a} = \infty$$

$$P\{T_b < \infty | S_0 = i\} = \lim_{a \rightarrow -\infty} \frac{(q/p)^{i-a} - 1}{(q/p)^{b-a} - 1} = \lim_{a \rightarrow -\infty} \frac{(q/p)^{i-a}}{(q/p)^{b-a}} = \frac{(q/p)^i}{(q/p)^b} = (q/p)^{i-b}$$

- Note also that: $a \leq i \leq b \Rightarrow i - b \leq 0 \Rightarrow (q/p)^{i-b} < 1$.
- Hence: $0 < P\{T_b < \infty | S_0 = i\} = (q/p)^{i-b} < 1$
- **So, even though we move towards a , we still have a nonzero probability to hit b**

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- To summarize:
 - For **biased RWs**, when $a = -\infty$, the probability of hitting the boundary b depends on the p, q and is given by:

$$P\{T_b < \infty | S_0 = i\} = \begin{cases} 1, & p = q \\ 1, & p > q \\ 0 < (q/p)^{i-b} < 1, & p < q \end{cases}$$

